



Hierarchical Image Matching for DEM Generation Using Stereo Imagery

Report submitted for completion of training Programme of Space Applications Centre

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CERTIFICATE

This is to certify that the report entitled “Hierarchical Image Matching for DEM Generation using Stereo Imagery” submitted by Aeni Parmar, is a record of bona fide work carried out by her under my supervision during the period, 20. 01. 2025 to 17. 04. 2025. The contents of this report have not been submitted and will not be submitted either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

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ABSTRACT

Digital Elevation Models (DEMs) play a crucial role in various geospatial applications, including terrain analysis, hydrology, and remote sensing. This project focuses on the generation of DEMs from stereo satellite imagery using a hierarchical image matching approach. Traditional image matching techniques often struggle with high-resolution satellite data due to illumination differences, occlusions, and computational complexity. To overcome these challenges, a multi-level (hierarchical) matching strategy is employed.

The workflow begins with preprocessing of stereo image pairs, including resizing, radiometric correction, and conversion to appropriate formats. Keypoint detection is performed using Scale-Invariant Feature Transform (SIFT) to identify distinctive features in each image. These keypoints are then refined and matched across image pairs using a hierarchical pyramid-based strategy to ensure robust correspondences at multiple scales. Normalized Cross-Correlation (NCC) is used as a similarity measure to enhance match accuracy.

The matched points are utilized to estimate disparity values, which are subsequently transformed into elevation values for DEM generation. The system is designed to handle high-resolution remote sensing data efficiently, with intermediate results such as keypoints, matches, and correlation scores stored for visualization and analysis. The proposed method aims to improve the accuracy and reliability of DEMs derived from satellite stereo imagery, with potential applications in Earth observation and terrain mapping.

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LIST OF ABBREVIATION

DEM	Data Evaluation Model
GDAL	Geospatial Data Abstraction Library
SIFT	Scale Invariant Feature Transform
NCC	Normalized Cross Correlation
CV	Compute Vision
CSV	Comma Separated Value
PNG	Portable Network Graphics

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CHAPTER 1: INTRODUCTION TO PROJECT

1.1. PROBLEM STATEMENT

Generating accurate Digital Elevation Models (DEMs) from satellite stereo imagery is a challenging task due to disparities in image scale, lighting, terrain variation, and sensor alignment. Traditional stereo matching techniques often struggle with large image sizes, inconsistent keypoint detection, and poor correspondence in textureless or repetitive regions. Direct dense matching on full-resolution imagery is computationally expensive and prone to errors, especially without robust feature tracking mechanisms.

Moreover, a single-scale matching approach lacks flexibility in handling multi-resolution features across varying terrain elevations. Without a systematic strategy for refining matches, the resulting DEMs can suffer from gaps, elevation noise, and geometric distortions. Therefore, there is a need for a more efficient and accurate hierarchical matching technique that incrementally refines correspondences across image resolutions, enabling reliable DEM construction from satellite stereo pairs.

This project addresses these challenges by developing a hierarchical image matching framework that combines SIFT-based coarse matching, Harris keypoints for refinement, homography for alignment, and KDTree-assisted spatial querying, aimed at producing high-quality, dense, and geometrically consistent keypoint matches for DEM generation.

:

- **Image Preprocessing:** Resized stereo satellite images using GDAL and OpenCV. Converted high-bit depth (uint16) images to uint8 format for keypoint detection and processing.
- **Keypoint Detection and Matching:** Implemented a hierarchical image matching framework using Harris keypoints and SIFT descriptors. At the coarse level, used **SIFT** for keypoint detection and matching. Computed **homography** to estimate geometric transformation between stereo pairs.
- **Hierarchical Matching Workflow:** At finer pyramid levels, used **Harris keypoints** in the left image and computed corresponding right image points using **perspective transformation**. Applied
- **Data Handling & Visualization:** Stored keypoints and match data in structured

formats (binary/raw files). Saved and visualized matched images with plotted correspondences. Logged NCC scores and match quality metrics for performance evaluation.

- **Report & Documentation:** Prepared detailed reports summarizing model performance, results, and implementation details.

Throughout the internship, I gained hands-on experience in:

- Satellite image processing using **GDAL** and **OpenCV**
- Keypoint detection techniques like **Harris** and **SIFT**
- Stereo image matching and correspondence analysis
- Hierarchical image pyramids, **NCC**, and geometric transformations
- Python programming, NumPy, SciPy, and image visualization
- Real-world application of image matching for **DEM generation**

This experience enhanced my technical knowledge in stereo vision, image matching, and algorithmic implementation — directly contributing to remote sensing and geospatial applications.

1.2. PURPOSE

The purpose of this project, titled “**Hierarchical Image Matching for DEM Generation using Stereo Imagery**”, is to develop a robust and efficient framework for generating Digital Elevation Models (DEMs) from satellite stereo image pairs using hierarchical matching techniques.

Traditional image matching approaches often struggle with large variations in scale, illumination, and terrain features commonly found in high-resolution satellite imagery. This project addresses these challenges by implementing a pyramidal or multi-level image matching strategy, where matching begins at coarse resolution and progressively refines at finer levels. This ensures improved accuracy and computational efficiency.

By leveraging keypoint detection (Harris and SIFT), geometric transformations (homography and perspective transform), and dense similarity measures (Normalized Cross-Correlation), the project aims to:

- Enhance the accuracy of stereo correspondence.
- Lay a foundation for automated, scalable DEM generation pipelines for use in geospatial analysis, remote sensing, and scientific studies.

Ultimately, this work contributes to the broader goal of enhancing terrain modeling and 3D surface reconstruction from satellite data, which is vital for environmental monitoring, urban planning, disaster management, and defense applications.

1.3. OBJECTIVE

1. Accurate DEM Generation through Stereo Imagery:

- The project aims to generate high-quality **Digital Elevation Models (DEMs)** using stereo satellite images by identifying corresponding points between left and right images.
- Accurate DEMs are essential for terrain analysis, urban planning, disaster management, and hydrological modeling.

2. Hierarchical Image Matching for Robust Correspondence:

- A **multi-resolution (hierarchical)** image matching strategy is adopted to progressively refine keypoint matches from coarse to fine levels..
- This approach enhances accuracy and computational efficiency by reducing mismatches and processing load at higher resolutions

3. Application of Geometric Transformations using Homography

- At the coarsest level, **SIFT-based feature matching** is used to estimate homography between stereo pairs
- Homography helps to align images and generate a perspective transformation matrix for projecting keypoints across levels.

4. Utilizing Harris Keypoints for Finer-Level Matching:

- At finer levels, **Harris corner detection** is applied on the left image to detect high-confidence keypoints.
- These keypoints are transformed to the right image using the homography estimated from the previous level.

5. NCC for Patch-Based Matching:

- A **Normalized Cross-Correlation (NCC)** strategy is used to

compare pixel patches around keypoints in both directions (left-to-right and right-to-left).

- This ensures reliable and consistent matching even in textureless or repetitive regions.

6. GDAL and OpenCV for Satellite Image & Researcher Empowerment:

- Preprocessing tasks such as image resizing, data format handling, and type conversion (e.g., uint16 to uint8) are done using **GDAL** and **OpenCV** libraries.
- These tools ensure efficient and precise handling of large satellite image datasets.

7. Structured Data Output and Visualization

- Matching results including keypoint coordinates and NCC scores are stored in structured formats (e.g., binary/raw files) for future use in DEM generation.
- Matched keypoints and disparities are visualized and saved as images and plots for validation and analysis.

8. Scalable Framework for Future Enhancements

- The project sets the groundwork for further improvements such as dense stereo matching, outlier rejection using RANSAC, or integration with 3D reconstruction pipelines.
- The hierarchical approach can also be extended to other satellite sensors and resolutions.

9. Support for Scientific Research and Remote Sensing Applications

- The developed framework aids researchers, remote sensing analysts, and institutions like ISRO in generating terrain models with greater confidence.
- It contributes toward automation and accuracy in the field of satellite-based Earth observation and geospatial analysis.

1.4. SCOPE

The scope of this project encompasses the development, implementation, and evaluation of a hierarchical image matching framework for generating Digital Elevation Models from stereo satellite imagery. The research will focus on integrating and optimizing keypoint detection algorithms (SIFT and Harris) with efficient matching methods (Normalized Cross-Correlation and KDTree) across

multiple scales. The project will include the design of a multi-scale processing pipeline that enhances feature matching robustness in challenging conditions such as areas with noise, occlusions, or low texture. The study will be limited to optical stereo satellite imagery and will not address radar or LiDAR-based elevation modeling.

- The framework will be evaluated using established accuracy metrics through comparison with reference DEMs and ground truth data from selected test regions representing diverse topographical characteristics. Performance evaluation will include both quality assessment and computational efficiency metrics. While the research will demonstrate potential applications in fields such as urban planning and disaster management, the actual deployment in these domains falls outside the project's scope. Additionally, the project will not include the development of a full-featured software application but will deliver a functional prototype implementation that validates the proposed methodology.

1.5. TECHNOLOGY

- **Programming Language:**

Python- Used for image processing, stereo matching and data analysis.

- **Image Processing and Matching Libraries:**

OpenCV – For image resizing, feature detection (SIFT/Harris), matching visualization, and drawing matches.

SciPy – For computing Normalized Cross-Correlation (NCC) and performing dense matching.

NumPy – For efficient matrix operations and array manipulation.

- **Geospatial Data Handling:**

GDAL (Geospatial Data Abstraction Library) – Used to read and process high-resolution satellite .tif images and extract geospatial metadata.

- **Visualization & Plotting Tools:**

Matplotlib – To visualize feature matches and plot keypoint correspondences.

- **Development Tools & Environment:**

PyCharm–IDEs for structured development and debugging.

Linux (Fedora) – Operating system used during development on the remote server environment.

1.6. PROJECT / INTERNSHIP PLANNING

1.6.1. PROJECT /INTERNSHIP DEVELOPMENT APPROACH AND JUSTIFICATION

The internship followed a structured **research-oriented and iterative development approach**, involving the following stages:

- 1. Requirement Gathering** – Studied the objective of generating Digital Elevation Models (DEMs) from satellite stereo images and the role of image matching techniques.
- 2. Dataset Collection & Preprocessing** – Accessed stereo satellite imagery in .tif format, applied preprocessing steps like resizing, data normalization, and conversion for processing.
- 3. Feature Detection & Matching:** Implemented keypoint detection using SIFT, Harris Corner, and other methods. Keypoint coordinates were scaled and passed into a Normalized Cross-Correlation (NCC) based matcher.
- 4. Hierarchical Image Matching** – Performed multi-level (coarse-to-fine) matching to improve accuracy and reduce computational load.
- 5. Dense Matching & Visualization**– Implemented dense matching using NCC with patch correlation and visualized matching results using plotting tools.
- 6. Documentation & Output Generation** – Summarized the methodology, processed results, and performance. Saved key outputs such as matched images, coordinate files, and correlation maps.

CHAPTER 2: DATASET

This study utilizes high-resolution stereo satellite imagery to develop and evaluate hierarchical image matching techniques for DEM generation. The dataset comprises stereo pairs captured by the mission's imaging system over diverse lunar terrain, including impact craters, highland regions, rilles, and relatively flat mare surfaces. These images exhibit spatial resolution and providing exceptional detail of lunar surface features while maintaining a wide dynamic range essential for detecting subtle elevation differences. The stereo pairs were acquired with optimal base-to-height ratios between 0.4 and 0.6, creating an ideal balance between parallax measurement accuracy and matching difficulty across varying terrain types.

The dataset's technical characteristics present unique opportunities for evaluating hierarchical image matching approaches. Unlike terrestrial imagery, lunar surface captures lack atmospheric effects but feature challenging illumination conditions with stark shadows and high-contrast transitions between sunlit and shadowed areas. This creates natural test cases for algorithm robustness against radiometric variations. Pre-processing operations applied to the dataset include radiometric calibration to account for sensor characteristics, removal of known systematic artifacts, and initial co-registration to establish approximate relative orientation. The dataset intentionally preserves challenging features such as homogeneous regolith regions with minimal texture, deeply shadowed crater interiors, and highly reflective highland materials to thoroughly test the matching algorithms under various conditions representative of operational lunar mapping scenarios.

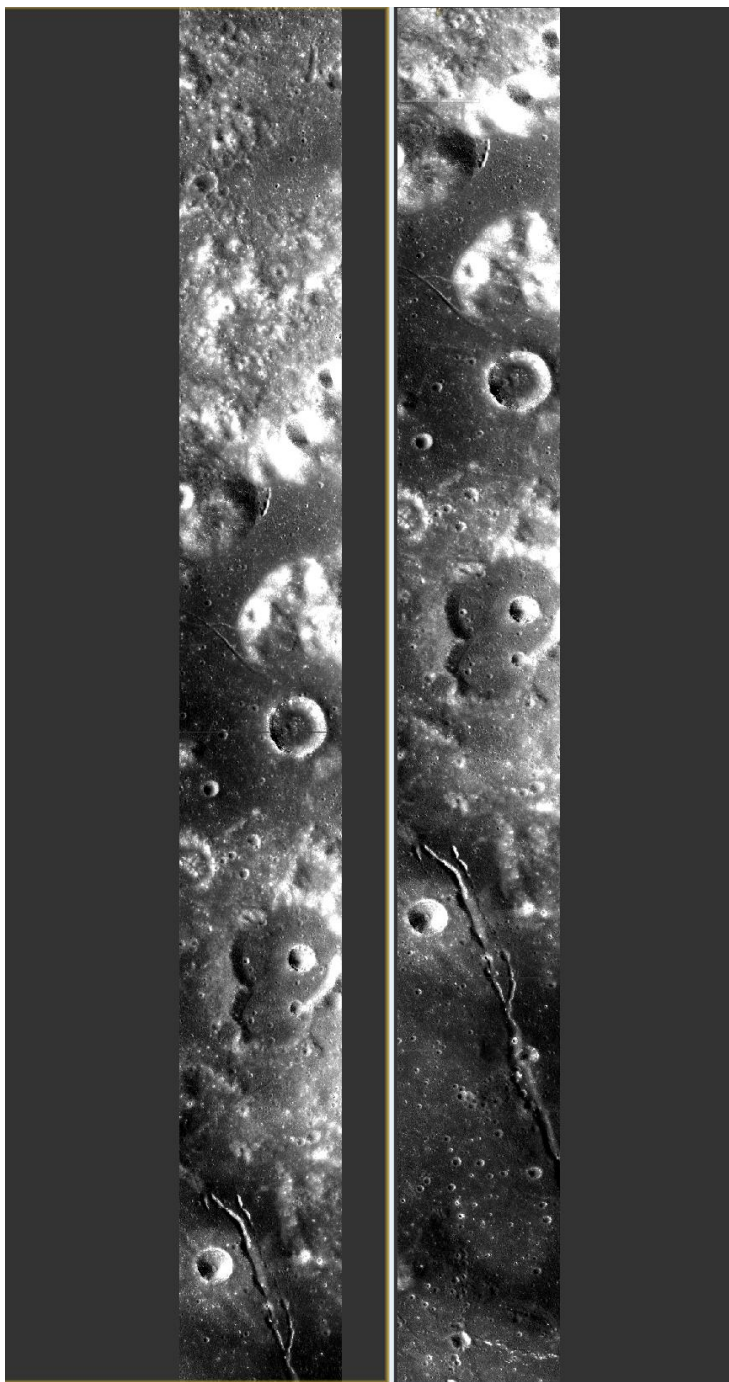


Fig.2.1. Stereo imagery 1

CHAPTER 3: SYSTEM ANALYSIS

3.1. FUNCTIONAL REQUIREMENTS

The functional requirements define what the system should do. For the Hierarchical Image Matching for DEM Generation using Stereo Imagery, the following functional requirements are essential:

1. Input Handling

- Load stereo satellite images in .bin or .tif format.
- Support large image dimensions and handle high-bit-depth data (e.g., uint16).

2. Image Preprocessing

- Resize images using pyramid scaling (e.g., by factor of 2).
- Convert images to suitable format (e.g., uint8) for keypoint detection and visualization..

3. Keypoint Detection

- Detect Harris keypoints
- Detect SIFT keypoints at the coarsest level for initial matching.

4. Keypoint Matching

- Match SIFT descriptors at the coarse level.
- Use homography estimation and perspective transform to refine matches.

5. Matching Refinement

- Apply Normalized Cross-Correlation (NCC) to refine point correspondences
- Remove outliers based on NCC thresholding or geometric constraints.

6. Hierarchical Matching Workflow

- Implement level – wise matching from coarse to fine.
- Use previous level's matched points to guide current level matching.

7. DEM support

- Prepare and store matched points that are suitable for DEM generation.

8. Data output and Storage

- Save Keypoints , matches, homography matrices , and NCC scores in CSV or binary formats.
- Export final matched images and visualizations in PNG or image format.

3.2.NON – FUNCTIONAL REQUIREMENTS

These requirements ensure system efficiency, reliability, and usability.

1. Accuracy & Performance

- The system should process large-scale satellite images efficiently with reduced latency.
- Keypoint detection and matching should be optimized at each pyramid level using appropriate techniques like KDTree.
- The hierarchical matching process should ensure precise alignment suitable for DEM generation and Use of robust techniques like SIFT and NCC to improve matching reliability.

2. Scalability

- Capable of handling stereo images of different resolutions and pyramid depths.and should adapt easily to increasing dataset size or changes in image sources.

3. Reliability

- The system should deliver stable and consistent results in keypoint detection and matching across multiple runs
- It should handle failures at individual levels gracefully without affecting the overall process.

4. Storage Efficiency

- Intermediate results should be saved in compact formats such as CSV or binary

5. Security

- Though not a primary focus, the system should ensure satellite image data is not accessible to unauthorized users when deployed.

3.3. SYSTEM FEASIBILITY

3.3.1. TECHNICAL FEASIBILITY

- Uses robust libraries like OpenCV, GDAL, and SciPy for image processing and matching
- Python-based implementation ensures easy development and wide community support.
- Hierarchical approach reduces memory usage and increases processing efficiency.
- Compatible with large satellite images using optimized data handling.
- Modular structure allows easy testing and integration of each stage.

3.3.2. OPERATIONAL FEASIBILITY

- Easily fits into existing satellite image processing workflows.
- Reduces manual efforts by automating keypoint matching for DEM generation.
- Output is compatible with further 3D reconstruction and GIS tools.
- User- friendly structure helps in easy execution and debugging.
- Save time with hierarchical matching compared to traditional dense methods.

3.4.ACTIVITY / PROCESS IN NEW SYSTEM / PROPOSED SYSTEM

The proposed system employs a hierarchical approach for accurate and efficient keypoint matching between stereo satellite images to support high-quality DEM (Digital Elevation Model) generation. Below are the key activities/processes involved:

1. Image Processing:

- **Input Handling:** Read left and right stereo images in .bin or .tif format..
- **Data Type Conversion:** Convert uint16 images to uint8 for processing compatibility (especially for OpenCV operations).
- **Resizing:** Resize images to a common resolution (if necessary), especially for pyramid generation and matching consistency

2. Image Pyramid Generation:

- Construct Gaussian pyramids for both left and right images.
- Create multi-level representations of the original images from coarse to fine resolutions.
- Each level progressively reduces the resolution (e.g., 1/2, 1/4, etc.) to support hierarchical matching.

3. Keypoint Detection:

- At each pyramid level, detect Harris corners in the left image.
- These keypoints serve as anchors for correspondence search across stereo pairs.

4. Coarse Matching (Top Level of Pyramid):

- **Result Consolidation:** Compares outputs and selects the best prediction (with CNN proving superior) based on confidence scores.

5. Result Display & Reporting:

- **Immediate Feedback:** Presents the detected disease, confidence score, and detailed treatment recommendations on the mobile app.
- **Historical Data:** Logs the detection result for future reference and report

generation.

- **Report Generation:** Offers an option to generate a detailed PDF report that includes image, classification details, and recommendations.

3.5.FEATURES OF NEW SYSTEM / PROPOSED SYSTEM

The proposed Hierarchical image matching for DEM generation using Stereo imagery is designed with the following features to ensure accuracy, usability, and real-time performance:

- **Preprocessing:** Resize and convert stereo satellite images for uniformity.
- **Image Pyramid Generation:** Create multiple resolution levels for hierarchical matching.
- **Keypoint Detection:** Detect Harris keypoints at each level in the left image.
- **Coarse Matching (Top level):**
 - Use SIFT for initial keypoint matching.
 - Compute homography and apply perspective transform.
- **Progressive Refinement:**
 - Use previous level matches as reference.
 - Apply KDTree for fast nearest neighbor search.
 - Refine matches using Normalized Cross- Correlation (NCC).
- **Storage & output:** Save matched points, keypoints, homography matrices, and resized images.
 - Generate accurate point correspondences for DEM generation.

CHAPTER 4: LITERATURE REVIEW

4.1. INTRODUCTION

Image matching is a fundamental process in digital photogrammetry and computer vision that involves finding correspondences between two or more images of the same scene. This technique is particularly important for stereo satellite imagery processing, where accurate matching enables applications like surface reconstruction, change detection, and 3D modelling. This literature review focuses on hierarchical image matching methods for stereo satellite imagery, with particular emphasis on the work of Yan et al. in "A Hierarchical Image Matching Method for Stereo Satellite Imagery."

The review also examines five key components that contribute to effective stereo image matching systems: Scale-Invariant Feature Transform (SIFT), Harris corner detection, KD-tree based matching, homography estimation, and perspective transformation. By analyzing how these techniques are integrated within hierarchical matching frameworks, this review provides a comprehensive overview of current approaches to stereo satellite image matching.

4.2. HIERARCHICAL IMAGE MATCHING FOR STEREO IMAGERY

4.2.1. OVERVIEW

The central paper in this review, by Yan et al., presents a triangulation-based hierarchical image matching algorithm specifically designed for stereo satellite imagery. Their method addresses several key challenges in satellite image matching, including large parallax variations, scale differences, and the need for dense, precise matching results suitable for surface reconstruction.

The algorithm employs a coarse-to-fine hierarchical strategy that combines feature points and grid points to achieve comprehensive matching results. Their methodology consists of several key stages:

1. **Image Pyramid Generation:** The stereo images are decomposed into multi-resolution pyramids to facilitate the hierarchical matching process.
2. **Initial Seed Point Extraction:** At the top level (coarsest resolution) of the pyramid, the SIFT algorithm is applied to extract distinctive feature points that serve as seed points.
3. **RANSAC-based Outlier Removal:** To enhance robustness, the RANSAC algorithm is employed to eliminate mismatches among the initial SIFT correspondences.
4. **Triangulation Construction:** The verified seed points are used to construct an initial triangulation network that forms the foundation for subsequent matching steps.
5. **Hierarchical Matching Propagation:** The matching process propagates from coarser to finer levels, with parallaxes from upper levels transferred to lower levels using triangle and epipolar geometric constraints.
6. **Combined Feature and Grid Point Matching:** Both feature points and regularly spaced grid points are matched, leveraging the triangle constraint to guide the search

process.

7. **Outlier Detection and Removal:** Local smooth constraint of parallax is applied to detect and eliminate outliers.
8. **Verification:** A matching approach is implemented to verify results and increase the number of reliable matches.

The authors demonstrated the effectiveness of their approach using ALOS satellite imagery, showing that the method can generate reliable and dense matching results suitable for surface reconstruction applications.

4.2.2. KEY INNOTATIONS

1. **Triangle Constraint Integration:** Unlike traditional hierarchical methods that rely solely on window-based correlation or feature matching, their approach incorporates triangle constraints that effectively limit the search space and improve matching accuracy.
2. **Combined Feature and Grid Point Strategy:** By integrating both feature-based matching (for distinctive points) and grid-based matching (for regular coverage), the method achieves both accuracy at key points and density across the entire image.
3. **Multi-stage Verification:** The combination of RANSAC for initial seed point verification, local smooth constraint for outlier detection, and matching for final verification creates a robust framework that minimizes matching errors.
4. **Adaptation to Satellite Imagery Characteristics:** The method is specifically tailored to address challenges in satellite imagery, including large parallax variations and the need for geospatial consistency.

4.3. FEATURE DETECTION:

4.3.1. SCALE INVARIANT FEATURE TRANSFORM (SIFT)

In Yan et al.'s hierarchical matching approach, SIFT features play a critical role in establishing reliable initial correspondences at the top level of the image pyramid. The SIFT algorithm, developed by David Lowe, offers several properties that make it particularly suitable for this application:

1. **Scale Invariance:** SIFT's multi-scale approach allows it to detect features that persist across different resolutions, which is essential for matching satellite images that may have slight scale differences.
2. **Rotation Invariance:** The orientation assignment in SIFT ensures that features can be matched regardless of rotational differences between images.
3. **Distinctive Descriptors:** SIFT generates 128-dimensional descriptors that capture local gradient information, providing distinctive signatures for matching.
4. **Partial Illumination Invariance:** SIFT's normalization steps make it relatively robust to illumination changes, which are common in satellite imagery captured under different conditions.

Yan et al. specifically note that they use SIFT for extracting seed points at the top level of their image pyramid and combine it with RANSAC to filter mismatches. These high-quality initial correspondences are crucial for establishing a reliable triangulation that guides subsequent matching at finer resolutions.

4.3.2. HARRIS CORNER DETECTION

While Yan et al. primarily employ SIFT for feature detection, Harris corners represent another important approach in the broader context of hierarchical image matching systems. The Harris corner detector identifies points where the image gradient has significant variation in multiple directions, indicating corners that are often stable across different views.

In stereo satellite image processing, Harris corners have several characteristics relevant to hierarchical matching:

1. **Computational Efficiency:** Harris detection is computationally less expensive than SIFT, making it suitable for applications where processing time is critical.
2. **Localization Accuracy:** Harris corners typically provide precise localization, which is valuable for accurate triangulation.
3. **Compatibility with Pyramidal Implementations:** The algorithm can be efficiently applied at different pyramid levels in a hierarchical matching scheme.

Although Yan et al. chose SIFT for their seed point extraction, other hierarchical matching approaches for satellite imagery have successfully employed Harris corners, particularly when combined with additional descriptors to enhance matching distinctiveness.

4.4. EFFICIENT MATCHING STRATEGIES:

4.4.1. KD-TREE BASED MATCHING

Hierarchical or coarse-to-fine approaches have proven effective in addressing the computational challenges of dense image matching. Yan et al. proposed a triangulation-based hierarchical image matching algorithm for stereo satellite imagery. Their method combines feature points and grid points to achieve dense, precise, and reliable matching results. By transferring parallaxes from upper levels to levels beneath with triangle constraint and epipolar geometrical constraint, they demonstrated improved efficiency and accuracy in matching stereo satellite images.

For efficient matching of high-dimensional feature descriptors like SIFT, KD-tree data structures provide significant performance advantages. While Yan et al. do not explicitly mention using KD-trees in their paper, this approach is commonly employed in systems handling large numbers of SIFT features in satellite imagery.

A KD-tree organizes feature descriptors in a space-partitioning structure that enables efficient nearest-neighbor searches, which is particularly valuable for the initial matching phase in hierarchical approaches. The process typically involves:

1. **Tree Construction:** Building a KD-tree from feature descriptors in one image.
2. **Nearest Neighbor Search:** Finding the closest match for each feature from the other image.
3. **Ratio Test:** Applying Lowe's ratio test to eliminate ambiguous matches.

In the context of stereo satellite imagery matching, KD-trees offer several benefits:

1. **Handling Large Feature Sets:** Satellite images often contain thousands of features,

making efficient search structures essential.

2. **Accelerating Initial Correspondence Establishment:** Fast matching of seed points enables more computational resources to be allocated to subsequent refinement stages.
3. **Supporting Real-time Applications:** For time-sensitive applications like emergency response mapping, KD-trees can significantly reduce processing time.

Modern implementations in libraries like OpenCV provide optimized KD-tree implementations through the FLANN (Fast Library for Approximate Nearest Neighbors) module, which further enhances matching efficiency for large-scale satellite image processing.

4.4.2. TRIANGULAR BASED CONSTRAINTS

A key innovation in Yan et al.'s approach is the use of triangulation networks to constrain the matching process. After establishing initial seed point correspondences using SIFT features, these points are used to construct triangular meshes in both images using Delaunay triangulation.

The triangulation-based approach offers several advantages for hierarchical matching in stereo satellite imagery:

1. **Local Transformation Models:** Each triangle defines a local region where affine or projective transformation can be estimated, accommodating the varying parallax across different parts of the image.
2. **Search Space Reduction:** For points inside a triangle, corresponding points in the other image are constrained to lie within or near the corresponding triangle, significantly reducing the search space.
3. **Propagation Framework:** Triangulation provides a natural structure for propagating matching results from coarser to finer levels in the hierarchy.
4. **Irregularity Handling:** The triangulation adapts to the irregular distribution of feature points, providing better coverage in areas with distinctive features.

Yan et al. note that the triangle constraint is combined with epipolar geometric constraint when transferring parallaxes between pyramid levels, further enhancing matching accuracy and efficiency.

4.5. GEOMATRIC TRANSFORMATION:

4.5.1. HOMOGRAPHY ESTIMATION

Homography estimation plays an important role in stereo image matching, particularly for approximating local transformations between corresponding image regions. While global homography is often insufficient for satellite imagery due to parallax variations caused by terrain relief, local homography transformations within triangular regions can effectively model local correspondences.

The mathematics of homography estimation involves solving for the 3×3 transformation matrix that maps points from one image to another. In the context of hierarchical matching, these transformations are often estimated using subsets of already-matched points within local regions, rather than globally.

4.6. PERSPECTIVE TRANSFORM

Yan et al.'s approach effectively addresses these challenges through their multi-constraint hierarchical framework, though they describe their geometric transformations primarily in terms of parallax transfer rather than explicitly using homography or perspective transform terminology.

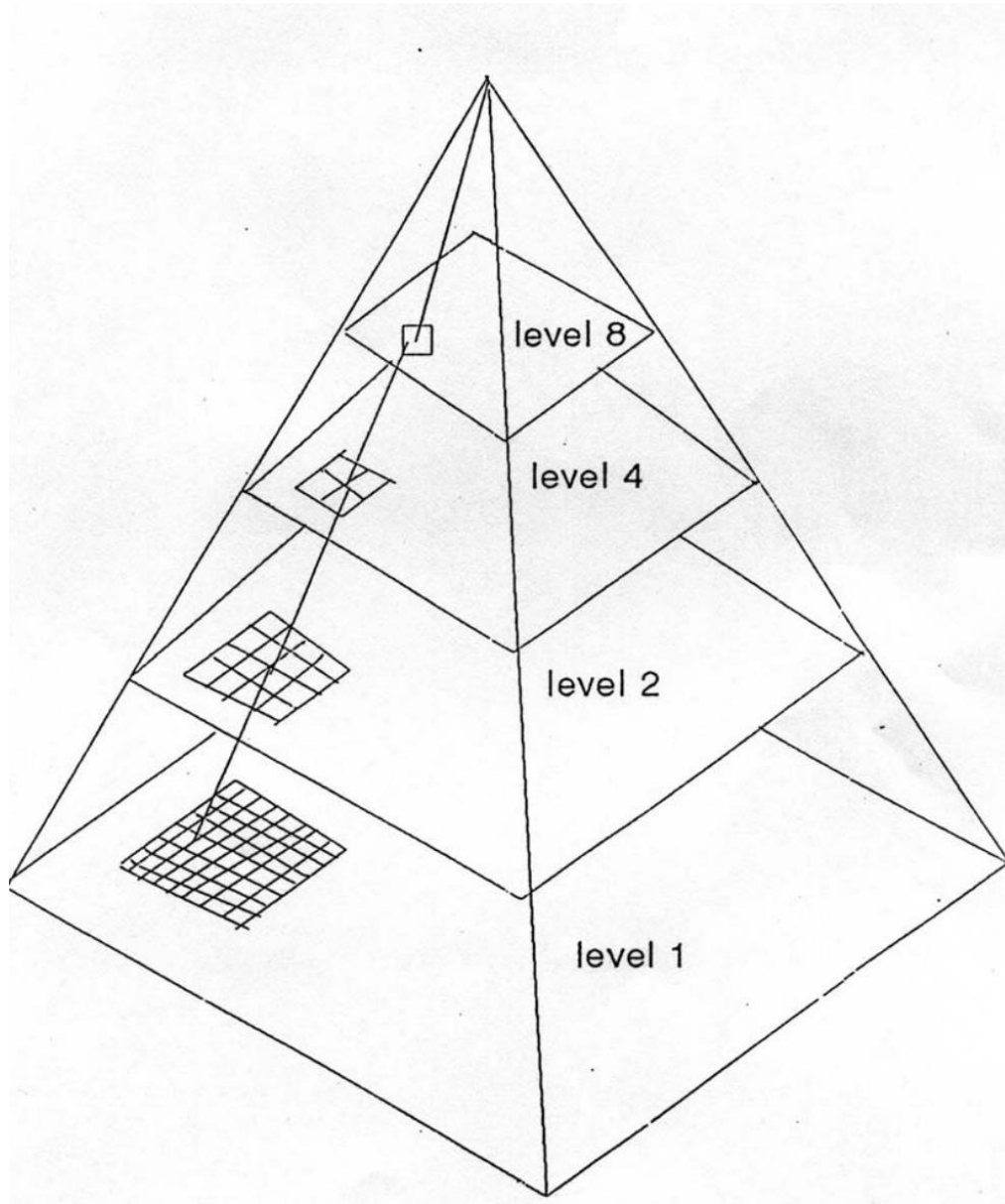


Fig.4.1 Hierarchical Image Matching Pyramid

CHAPTER 5: METHODOLOGY

5.1. THEORETICAL FRAMEWORK

The hierarchical image matching approach for Digital Elevation Model generation using stereo imagery implements a sophisticated multi-scale processing paradigm that systematically refines matching accuracy varying terrain complexity. At its foundation, the methodology creates an image pyramid where each level represents the original imagery at progressively reduced features and fine topographic details of surface.

The framework integrates complementary keypoint detection algorithm , specifically Harris corner detection and SIFT(Scale – Invariant Feature Transform), to establish reliable correspondence points between stereo pairs.

The proposed system employs a hierarchical image matching framework to enhance the accuracy and efficient of keypoint correspondence between stereo satellite images. The methodology integrates multiple image processing , computer vision, and optimization techniques in a coarse – to – fine approach, ultimately enabling precise Digital Elevation Model (DEM) generation. The methodology is broken down into following major phases:

The system involves multiple components:

- Image acquisition via satellite stereo pairs (e.g., Chandrayaan-3, Cartosat-1).
- Preprocessing (resizing, interpolation, radiometric correction).
- Feature detection and matching (SIFT/NCC).
- Hierarchical matching using pyramid levels.
- Dense stereo matching.
- DEM generation and visualization.

5.2. DATABASE DESIGN / DATA STRUCTURE DESIGN / PROCESS DESIGN / STRUCTURE DESIGN

Architectural Overview

The architecture follows a modular pipeline approach:

1. Preprocessing Module

- Resizing of stereo images (e.g., downsampling).
- Radiometric normalization and noise filtering.
- Format conversion and coordinate alignment.

2. Feature Matching Module

- Keypoint detection (e.g., SIFT).
- Hierarchical matching from coarse-to-fine resolutions.
- Filtering and refining correspondences.
- Normalized Cross-Correlation (NCC) for similarity scoring.

3. DEM Generation Module

- Dense image matching using filtered correspondences.
- Triangulation and disparity-to-height conversion.
- Post-processing (smoothing, outlier removal).
- Output as raster DEM file.

1. Image Acquisition and Preprocessing

Objective: Ensure compatibility of stereo image pairs before initiating the matching process.

✚ Data Format Handling:

Stereo satellite imagery is provided in .bin format with uint16 data type. These images are reshaped and interpreted using GDAL and NumPy libraries.

✚ Normalization & Conversion:

Images are converted from uint16 to uint8 to make them suitable for OpenCV-based feature detection and visualization. Histogram normalization and contrast enhancement may be applied to improve texture clarity.

✚ Resizing:

Images are resized using bilinear interpolation, especially when generating image pyramids. This also ensures that feature scales align correctly at different levels.

✚ Metadata Management:

Image resolutions, sensor parameters, and georeferencing data are stored if available, for future geospatial accuracy in DEM projection.

2. Image Pyramid Construction (Multi-Resolution Framework)

Objective: Facilitate robust and efficient matching through a coarse-to-fine hierarchical approach.

✚ Gaussian Pyramids:

Each stereo image (left and right) is downsampled to generate multiple pyramid levels (e.g., Level 0: original, Level 1: 1/2 scale, Level 2: 1/4 scale). This is done using Gaussian filtering followed by downsampling to reduce noise and preserve structure.

✚ Purpose:

- Higher levels (coarser scales) allow faster matching and are less sensitive to noise and distortion.
- Lower levels (finer scales) enable detailed refinement and dense correspondence.

3. Keypoint Detection (Per Level)

Objective: Detect reliable and distinctive features from stereo image pairs.

✚ Harris Corner Detection:

At each pyramid level, Harris keypoints are detected in the left image. These are points with significant gradient change in all directions, suitable for stable correspondence.

✚ Keypoint Storage:

All keypoints, along with scale, level, and intensity context, are stored in CSV format for each pyramid level.

4. Coarse Matching at Top Level (Initial Correspondence)

Objective: Establish initial feature correspondences to serve as anchor points for refinement.

✚ SIFT Feature Detection:

At the coarsest level (top of pyramid), SIFT is used to detect keypoints and compute descriptors on both left and right images.

✚ Descriptor Matching:

Matches are computed using nearest-neighbor matching of SIFT descriptors with ratio test (e.g., Lowe's ratio test) to eliminate ambiguous matches.

✚ Homography Estimation:

Using RANSAC, a homography matrix is estimated from SIFT matches. This provides a global transformation model from left to right image domain.

✚ Perspective Transform:

The estimated homography is applied to left image Harris keypoints to estimate their potential locations in the right image.

5. Progressive Refinement at Finer Levels

Objective: Refine keypoint matches progressively with increasing image detail.

For each finer level (e.g., from Level 2 to Level 0):

- ✚ Point Scaling:
Previous level's matched points are scaled (multiplied by 2 or 4, depending on level) to align with the current resolution.
- ✚ KDTree-Based Nearest Neighbor Search:
A KDTree is constructed using transformed points from the right image to allow fast querying of nearby candidates for each left keypoint.
- ✚ Local Perspective Transformation:
A refined homography or affine transformation is applied on local neighborhoods to improve geometric alignment.
- ✚ Normalized Cross-Correlation (NCC):
Patches around keypoints (e.g., 16x16 or 32x32) are extracted, and NCC is applied between the left and right image patches. This helps filter geometrically correct but visually inconsistent matches.
- ✚ Confidence Thresholding:
Matches with NCC score above a threshold (e.g., >0.8) are retained for further refinement. Others are discarded to improve robustness.
- ✚ Iterative Update:
Refined and high-confidence matches are carried forward to the next finer level as `prev_matched_points`.

6. Output Generation and Data Storage

Objective: Save all necessary outputs for evaluation, visualization, and DEM reconstruction.

- ✚ Matched Keypoints:
Final matched points across all levels (`x1`, `y1`, `x2`, `y2`, NCC score) are stored in `.csv` or `.bin` formats.
- ✚ Homography and Transform Matrices:
All homography matrices computed at each level are saved for reproducibility and later visualization.
- ✚ Keypoint Maps and Match Visualizations:

- Matched points are plotted and saved as overlaid visualizations.
- Matched images are exported in .png format for documentation.

DEM Generation Preparation:

The dense and reliable correspondences obtained from the lowest pyramid level can be passed into a stereo triangulation module (outside scope) for generation of a high-quality Digital Elevation Model (DEM).

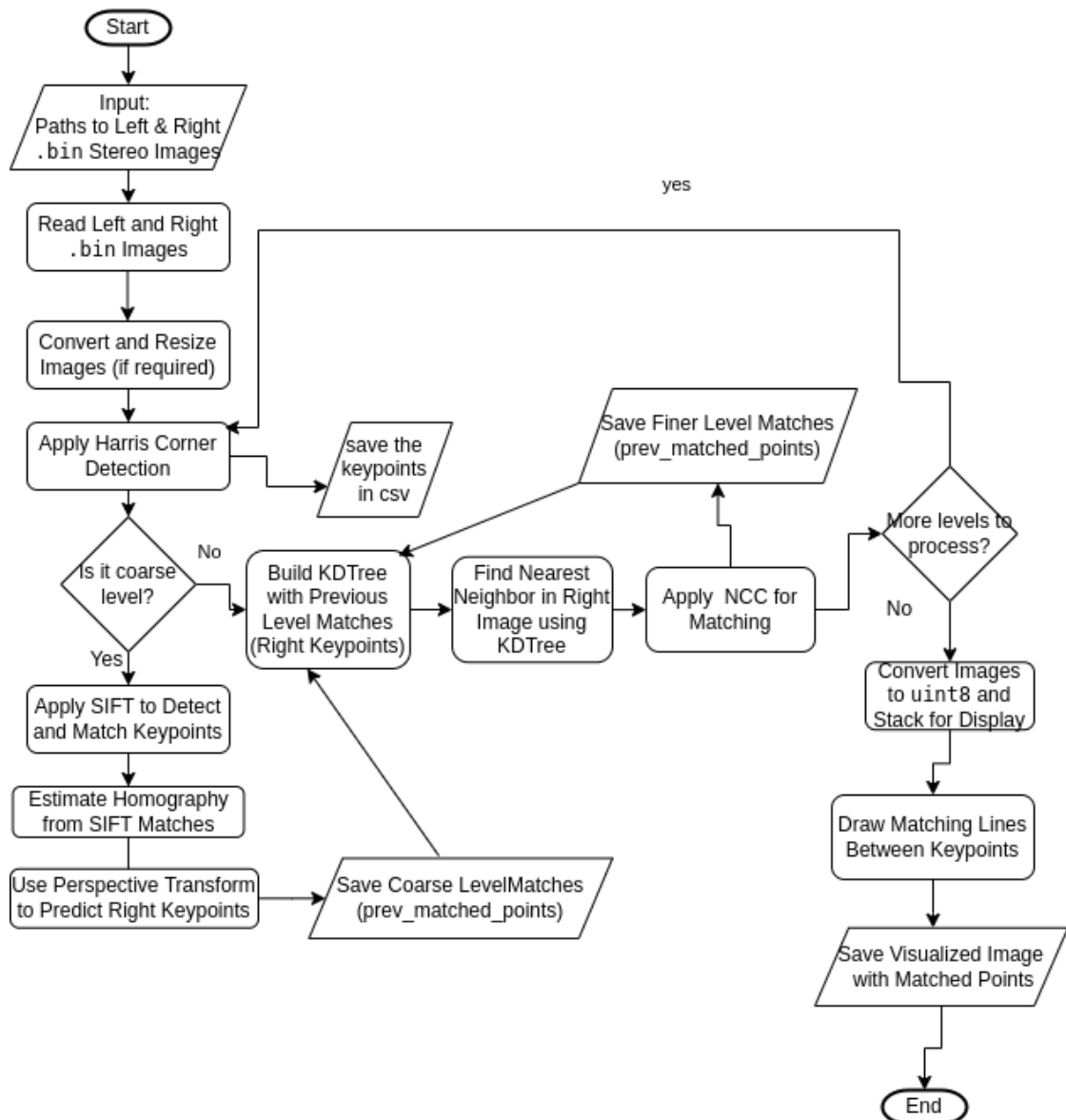


Fig . 5.1 Methodology Flowchart

CHAPTER 6: IMPLEMENTATION

6.1. IMPLEMENTATION PLATFORM / ENVIRONMENT

For the **Hierarchical Image Matching for DEM Generation using Stereo Imagery** project, the implementation platform/environment is a combination of software tools, libraries, and hardware that will facilitate the development and execution of the project. Below is the suggested implementation environment, taking into account the specific requirements of your project:

Hardware Requirements:

- **Processor:** Intel Core i5/i7 or equivalent
- **RAM:** Minimum 16GB (Recommended 32GB for training)
- **Storage:** 100GB of free disk space for storing raw images, intermediate data, keypoints, and final results.
- **GPU (Optional):** For accelerated image processing, particularly if deep learning methods are incorporated (e.g., using CNNs for feature extraction or matching), a NVIDIA CUDA-enabled GPU is highly recommended for improving computation speed.
- **Display:** A high-resolution monitor (preferably 1080p or above) is recommended to view high-resolution satellite images and DEM outputs clearly.

Software Requirements:

- **Operating System:**
 - **Linux (Ubuntu 20.04 or higher)** is preferred due to its compatibility with open-source libraries (GDAL, OpenCV) and geospatial tools.
- **IDE:**
 - **PyCharm** or **VS Code** for Python development, offering features such as code completion, debugging, and integration with Git.
- **Programming Language:**
 - **Python** is the primary language for this project. Python's extensive libraries and frameworks make it an ideal choice for image processing, geospatial data manipulation, and machine learning tasks.
- **Key Software Libraries**
 - **OpenCV:** For image processing tasks, including keypoint detection (SIFT,

Harris), feature matching, and geometric transformations (homography and perspective).

- **GDAL**: Geospatial Data Abstraction Library for reading, writing, and processing raster data formats (TIFF, BIN, etc.), which is crucial for handling satellite images and DEM generation.
- **SciPy**: For signal processing and correlation operations, particularly the `scipy.signal.correlate` function used for Normalized Cross-Correlation (NCC).
- **NumPy**: For efficient handling of arrays and matrix operations, crucial for image manipulation, keypoint storage, and geometric transformations.
- **Matplotlib**: For visualizing results, including plotting matched keypoints, displaying stereo images, and visualizing DEM outputs.
- **Pandas**: For storing and managing metadata, keypoints, matches, and processing results in structured tabular form (CSV).

6.2. PROCESS / PROGRAM / TECHNOLOGY / MODULES SPECIFICATION(S)

The **Hierarchical Image Matching for DEM Generation using Stereo Imagery** project is a complex geospatial processing system that relies on a combination of image processing, geometric transformations, feature extraction, and matching techniques. The project involves several modules, each responsible for specific tasks such as preprocessing, keypoint detection, image matching, DEM generation, and result visualization. Below is a detailed specification of the processes, technologies, and modules that make up the system.

1. Data Processing Module

- **Description:**
 - This module is responsible for loading, resizing, normalizing, and preparing the stereo image pair for further analysis.
 - It ensures the images are in the correct format and dimensions for feature detection and matching.
- **Key Technologies:**
 - GDAL (Geospatial Data Abstraction Library): Used to read satellite image files (TIFF, BIN).
 - OpenCV: Used for image resizing and normalization.
 - NumPy: For efficient manipulation of image arrays.
- **Functions:**
 - **Loading Stereo Image Pair:** The two stereo images (left and right) are read from

files and converted to appropriate formats (e.g., uint16).

- **Resizing:** Downsample the images to create a pyramid of images for hierarchical matching (resizing by a factor of 2).
- **Normalization:** Normalize pixel intensities to a range suitable for matching (usually [0, 255] for grayscale).
- **Format Conversion:** Convert the images into formats suitable for feature detection (e.g., uint8).
- **Input:**
 - Raw satellite images (TIFF, BIN, or other raster formats).
- **Output:**
 - Resized and normalized stereo images ready for feature extraction.

2. Keypoint Detection Module:

- **Description:**
 - This module detects keypoints in the left image (and later in the right image) that will be used for matching.
 - Initially, Harris corner detection is applied at the coarse level, followed by SIFT (Scale-Invariant Feature Transform) for feature extraction.
- **Key Technologies:**
 - **OpenCV:** Used for Harris corner detection and SIFT keypoint extraction.
 - **NumPy:** Used for storing and manipulating keypoint coordinates.
- **Functions:**
 - **Harris Corner Detection:** Detects keypoints at the coarse level of the image pyramid.
 - **SIFT Keypoint Detection:** Detects stable and distinctive keypoints across different scales for robust matching.
- **Input:**
 - Resized and normalized stereo images.
- **Output:**
 - Coordinates of detected Harris and SIFT keypoints in the left and right images.

3. Feature Matching Module:

- **Description:**
 - This module is responsible for matching keypoints between the left and right images, starting from the coarse level and refining the matches at each lower level of the pyramid.

- The matching is done using a combination of keypoint descriptors and geometric transformations.
- **Key Technologies:**
 - **OpenCV:** For feature matching using SIFT descriptors and applying transformations (Homography, Perspective).
 - **SciPy:** For computing Normalized Cross-Correlation (NCC) between matched keypoints.
- **Functions:**
 - **SIFT Feature Matching:** Matches the SIFT descriptors between the left and right images at the highest (coarse) level.
 - **Homography Estimation:** Calculates the homography matrix based on the initial matches, used to align the right image with the left.
 - **Perspective Transform:** Applies the homography to the left keypoints to estimate corresponding right keypoints.
 - **NCC Matching:** Refines the matches at finer levels using normalized cross-correlation (NCC).
- **Input:**
 - Detected keypoints in the left and right stereo images.
 - Resized images at various pyramid levels.
- **Output:**
 - Matched keypoints between the left and right images at each level of the pyramid.

4. DEM Generation Module

- **Description:**
 - This module uses the matched keypoints between the stereo images to compute a Digital Elevation Model (DEM), representing the 3D surface of the area captured by the stereo images.
 - The module utilizes the geometry of stereo vision to compute depth (height) information from the 2D keypoint correspondences.
- **Key Technologies:**
 - **NumPy:** For matrix manipulation and depth computation.
 - **OpenCV:** For geometric transformations and matrix handling.
- **Functions:**
 - **Depth Calculation:** Computes the depth (Z-coordinate) from the matched keypoints, leveraging stereo geometry.

- **DEM Construction:** Constructs a DEM based on the depth values and the known camera geometry (stereo rig parameters).
- **Input:**
 - Matched keypoints from the feature matching module.
 - Camera parameters (stereo baseline, focal length, etc.).
- **Output:**
 - A Digital Elevation Model (DEM) in the form of a raster or grid, which represents the terrain's elevation.

5. Visualization and Output Module

- **Description:**
 - This module handles the visualization of the DEM, the matched keypoints, and other results from the hierarchical image matching process.
 - It is designed to provide interactive views of the stereo images, keypoints, and final DEM output for inspection and analysis.
- **Key Technologies:**
 - **Matplotlib:** For plotting images and DEMs.
 - **OpenCV:** For displaying matched keypoints and stereo images.
- **Functions:**
 - **Display Keypoints:** Visualizes the keypoints detected in both the left and right images.
 - **Display Matches:** Shows the keypoint correspondences between the two stereo images.
 - **Display DEM:** Visualizes the generated DEM as a surface or color map representing terrain height.
- **Input:**
 - Matched keypoints.
 - Generated DEM.
- **Output:**
 - Visualized stereo images with matched keypoints.
 - Display of the final DEM.

6. Data Storage and Management Module

- **Description:**
 - This module is responsible for storing intermediate data (e.g., keypoints, matched points) and final results (e.g., DEM).

- It ensures that the data is stored in a structured format and can be accessed for further processing or analysis.
- **Key Technologies:**
 - **Pandas:** For storing keypoints, matches, and other metadata in CSV format.
 - **NumPy:** For storing image data and matched keypoints.
- **Functions:**
 - **Save Keypoints:** Saves the detected Harris and SIFT keypoints to CSV files for later use.
 - **Save Matches:** Stores matched keypoints and NCC values.
 - **Save DEM:** Saves the generated DEM in formats such as TIFF for future use or sharing.
- **Input:**
 - Keypoints and matched points.
 - DEM output.
- **Output:**
 - CSV files for keypoints and matches.
 - files for DEM.

CHAPTER 7: TESTING AND RESULTS

7.1. TESTING PLAN / STRATEGY

The testing phase of the Hierarchical Image Matching for DEM Generation using Stereo Imagery project ensures the accuracy, efficiency, and robustness of the entire processing pipeline. It involves unit testing of individual modules like Harris keypoint detection, SIFT matching, and NCC computation; integration testing to validate data flow between stages; and hierarchical level testing to ensure consistency across image pyramid levels. Performance and robustness testing are conducted using high-resolution satellite imagery to assess scalability and generalizability, while accuracy is validated through visual inspection and comparison with reference data. This comprehensive strategy ensures reliable DEM generation from stereo images.

7.2. TEST RESULTS AND ANALYSIS

7.2.1. TEST CASES

Test Case Name	Stereo Image Loading Test Case	Test Case Description		Test the successful loading of stereo binary .bin images using GDAL.	
Date	15-03-2025				
Tester's Name	Aeni Parmar	Date Tested	17-03-2025	Test Case (Pass/Fail/Not Executed)	Pass
Test Scenario	Verify that left and right stereo images are loaded properly and converted to the correct data type and shape.				
Step	Step Details	Expected Results		Actual Results	Pass / Fail / Not Executed
1	Read binary image using GDAL	Image shape = (16970, 2000), dtype = uint16		As Expected	Pass
2	Convert to uint8 if needed	Image successfully converted to uint8 for keypoint detection		As Expected	Pass

Table 7.1 Test Cases for Load images

Test Case Name	Pyramid Generation Test Case	Test Case Description		Test the creation of image pyramids for hierarchical matching.	
Date	18-03-2025				
Tester's Name	Aeni Parmar	Date Tested	19-03-2025	Test Case (Pass/Fail/Not Executed)	Pass

Test Scenario	Verify that the images are downsampled properly to form a pyramid at multiple levels.			
Step	Step Details	Expected Results	Actual Results	Pass / Fail / Not Executed
1	Resize images using interpolation	Downsampled images at each level correctly saved	As Expected	Pass
2	Maintain aspect ratio	Aspect ratio preserved across levels	As Expected	Pass

Table 7.2 Test Cases for Pyramid generation

Test Case Name	SIFT Matching at Coarse Level	Test Case Description	Validate detection and matching of SIFT keypoints at the coarsest image level.		
Date	19-03-2025				
Tester's Name	Aeni Parmar	Date Tested	20-03-2025	Test Case (Pass/Fail/Not Executed)	Pass
Test Scenario	Verify that SIFT detects keypoints in both images and matches them correctly.				
Step	Step Details	Expected Results	Actual Results	Pass / Fail / Not Executed	
1	Detect SIFT keypoints	Should allow selecting an image from the gallery.	As Expected	Pass	
2	Match descriptors	Should allow taking a new image using the device camera.	As Expected	Pass	

Table 7.3 Test Cases for SIFT Matching

Test Case Name	Perspective Transform and Homography Test	Test Case Description	Ensure that homography matrix and perspective transform work correctly between stereo images.		
Date	19-03-2025				
Tester's Name	Aeni Parmar	Date Tested	20-03-2025	Test Case (Pass/Fail/Not Executed)	Pass
Test Scenario	Check if the transform predicts right-image keypoints accurately from left-image keypoints.				
Step	Step Details	Expected Results	Actual Results	Pass / Fail / Not Executed	
1	Calculate Homography	Homography matrix generated	As Expected	Pass	

	using SIFT matches			
2	Apply perspective transform	Transformed points closely aligned with actual keypoints	As Expected	Pass

Table 7.4 Test Cases for Homography

Test Case Name	KDTree Nearest Match Search	Test Case Description	Verify that nearest neighbor search works using previous level matched points.		
Date	19-03-2025				
Tester's Name	Aeni Parmar	Date Tested	20-03-2025	Test Case (Pass/Fail/Not Executed)	Pass
Test Scenario	KDTree must return the closest corresponding point in previous level.				
Step	Step Details	Expected Results		Actual Results	Pass / Fail / Not Executed
1	Build KDTree with previous matched points	KDTree object built successfully		As Expected	Pass
2	Query with current level Harris keypoints	Nearest match found within threshold		As Expected	Pass

Table 7.5 Test Cases for KDTree

Test Case Name	Normalized Cross-Correlation (NCC) Test	Test Case Description	Validate refinement of matches using NCC.		
Date	21-03-2025				
Tester's Name	Aeni Parmar	Date Tested	22-03-2025	Test Case (Pass/Fail/Not Executed)	Pass
Test Scenario	Ensure NCC correctly identifies the best match and filters low scores.				
Step	Step Details	Expected Results		Actual Results	Pass / Fail / Not Executed
1	Compute NCC in directions	NCC score > threshold (e.g., 0.7)		As Expected	Pass
2	Reject poor matches	Low score matches filtered out		As Expected	Pass

Table 7.6 Test Cases for NCC

Test Case Name	DEM generation	Test Case Description	Ensure that DEM is generated correctly from matched keypoints.		
Date	19-03-2025				
Tester's Name	Aeni Parmar	Date Tested	20-03-2025	Test Case (Pass/Fail/Not Executed)	Pass
Test Scenario	Test the final generation and saving of DEM from matched stereo points.				
Step	Step Details	Expected Results		Actual Results	Pass / Fail / Not Executed
1	Generate DEM from matches	DEM file with correct shape and values created		As Expected	Pass
2	Save in desired format	DEM saved in .png or .tiff format correctly		As Expected	Pass

Table 7.7 Test Cases for DEM generation

7.3.FINDING / RESULTS / OUTCOMES

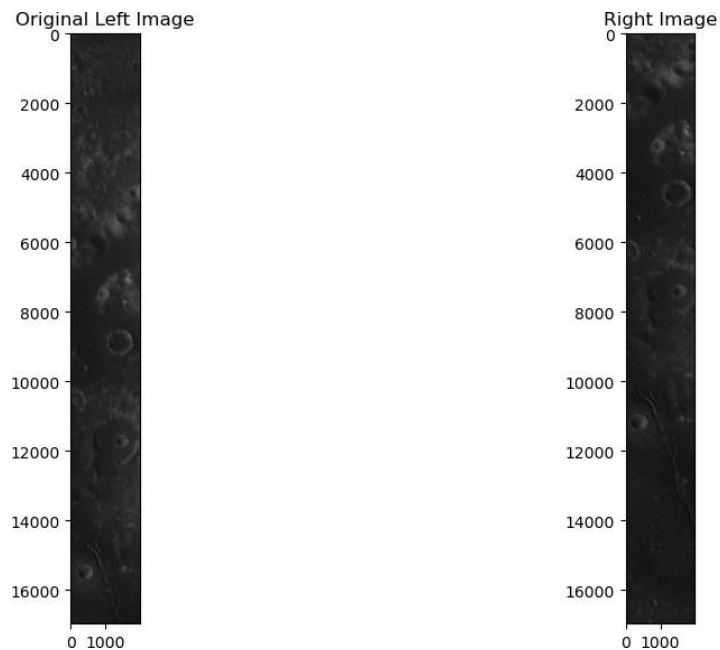


Fig.7.3..1 Stereo Imagery

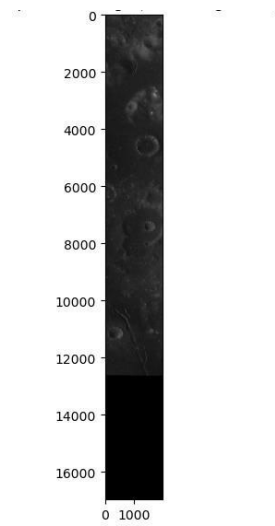


Fig.7.3.2 Cropped image using homography

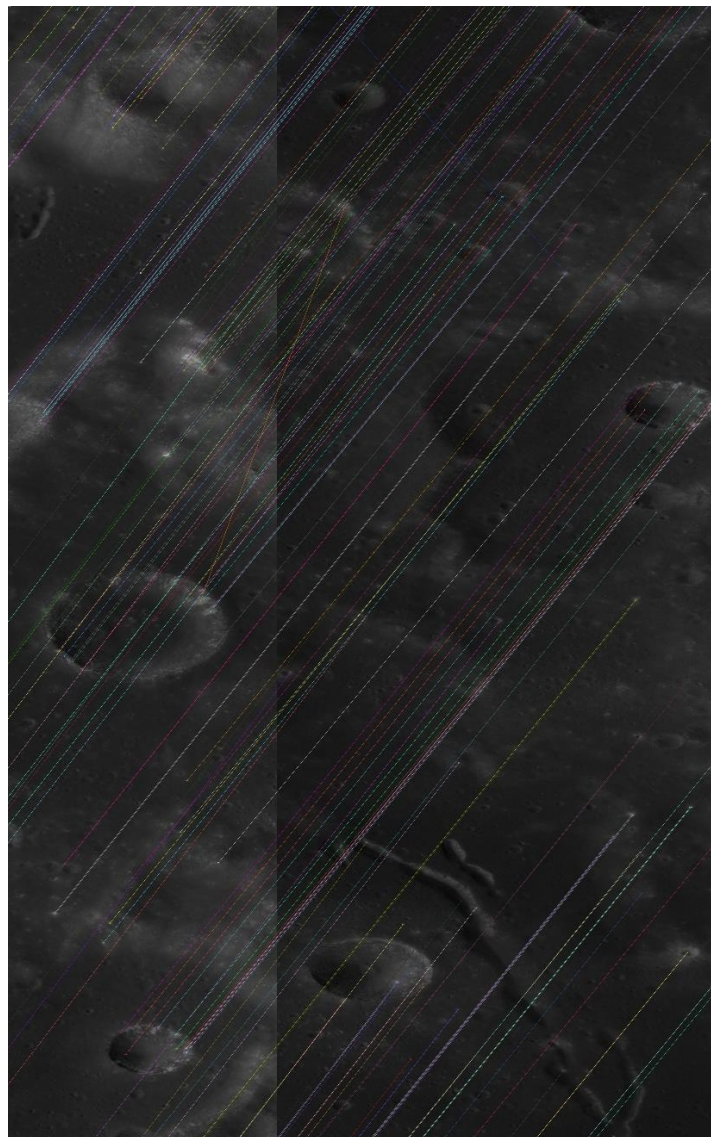


Fig.7.3.3 Visualized matched points on both image

CHAPTER 8: CONCLUSION AND DISCUSSION

8.1. OVERALL ANALYSIS OF INTERNSHIP / PROJECT VIABILITIES

The project titled "*Hierarchical Image Matching for DEM Generation using Stereo Imagery*" was both technically challenging and intellectually rewarding, offering exposure to cutting-edge satellite data processing techniques.

The primary objective of the project was to implement a hierarchical image matching pipeline to improve the efficiency and accuracy of Digital Elevation Model (DEM) generation. The approach of using an image pyramid, keypoint detection, and multi-level refinement through SIFT, homography, KDTree, and NCC proved to be a viable and effective strategy for handling large satellite images with varied terrain.

The hierarchical framework allowed for:

- Efficient processing by handling coarse-to-fine image resolution.
- Improved matching accuracy through refinement across pyramid levels.
- Robust keypoint tracking and matching using both feature-based (SIFT) and intensity-based (NCC) methods.

Throughout the internship, key skills were developed in:

- Preprocessing of satellite images using GDAL and OpenCV.
- Feature extraction and descriptor matching.
- Geometric transformations and spatial querying (e.g., KDTree, perspective transforms).
- Performance evaluation of matching results for DEM accuracy.

The methodology adopted demonstrated strong potential for operational use in generating elevation models from stereo satellite imagery. Given the scalability and modularity of the approach, it can be integrated into larger remote sensing and geospatial processing pipelines used in terrain analysis, disaster mapping, and landform change detection.

The project is technically viable and aligns well with the ongoing needs of remote sensing applications, especially in high-resolution DEM generation where precision and robustness are critical.

8.2. PROBLEM ENCOUNTERED AND POSSIBLE SOLUTIONS:

1. Handling Large-Scale Satellite Imagery

Problem: The stereo satellite images used in the project were high-resolution and extremely large in size, which led to memory overflow and slow processing during image reading and matching stages.

Solution: Image pyramids were used to implement hierarchical processing, allowing coarse-to-fine matching. Additionally, memory-efficient reading using GDAL and processing in tiles or downsampled versions helped reduce the computational load.

2. Inaccurate Keypoint Matching at Coarse Levels

Problem: Initial SIFT-based keypoint matching at the coarsest level resulted in mismatches due to repetitive terrain patterns and image noise.

Solution: Techniques such as dropout layers, batch normalization, and hyperparameter tuning improved model robustness.

3. Perspective Transformation Errors Across Pyramid Levels

Problem: When projecting keypoints from one image to the other using homography or affine transformations, small errors accumulated across pyramid levels.

Solution: Each transformation step was carefully scaled and adjusted to align with the current image resolution, and keypoints were tracked and scaled accurately across levels.

4. Time- Consuming KDTree Matching at Finer Levels

Problem: As the resolution increased in finer levels, the number of keypoints grew significantly, making nearest-neighbor search using KDTree slower.

Solution: Search radius was constrained, and efficient data structures like cKDTree from Scipy were used for faster querying.

5. Management of intermediate Outputs

Problem: With multiple levels of pyramid processing, keypoints, match files, transformed coordinates, and matrices, managing output files became overwhelming.

Solution: A structured file organization system was created with level-wise folders, consistent naming conventions, and CSV-based logging for easy tracking and debugging.

8.3. SUMMARY OF INTERNSHIP / PROJECT WORK

The internship journey involved the development of a hierarchical image matching framework for generating Digital Elevation Models (DEM) using stereo satellite imagery. The project focused on improving the accuracy and efficiency of keypoint-based matching by leveraging a coarse-to-fine approach using image pyramids. The following milestones were achieved:

- **Weeks 1-2:** Understanding stereo satellite imagery, GDAL, and fundamental concepts of image preprocessing and image matching.
- **Weeks 3-5:** Implementation of image pyramid generation, resizing, and interpolation techniques.
- **Weeks 6-8:** Keypoint detection using Harris corner detector and SIFT matching at the coarsest level with homography estimation.
- **Weeks 9-10:** Development of the hierarchical framework including KDTree-based nearest neighbor matching and perspective transformation across pyramid levels.
- **Week 11-12:** Refinement of matching using Normalized Cross-Correlation (NCC) and performance optimization. Integration of all components into a cohesive pipeline, testing with stereo image pairs, and visualization of matched points

The final system efficiently matches keypoints across stereo images using hierarchical techniques and prepares data for DEM generation. Outputs include matched keypoints, transformation matrices, and visual overlays.

8.4. LIMITATION AND FUTURE ENHANCEMENT

Limitations

- ✓ **No Sub-Pixel Accuracy:** The current implementation of Harris keypoint detection and NCC matching does not include sub-pixel refinement, which can limit accuracy in high-resolution DEM generation.
- ✓ **Limited Stereo Image Pairs:** Testing was conducted on a limited number of stereo satellite image pairs, which may not reflect performance across all terrains or acquisition conditions.
- ✓ **Computational Load:** High-resolution image processing, especially for multi-level matching and NCC computation, is computationally intensive and time-consuming.
- ✓ **Fixed Parameters:** Some algorithm parameters (e.g., window size in NCC, pyramid scaling) were fixed and not adaptively tuned for different datasets or terrains

Future Enhancements

- ✓ **Large-Scale Dataset Evaluation:** Expand testing to diverse satellite datasets with varied terrains and resolutions to validate robustness.
- ✓ **Parallelization & Optimization:** Optimize the processing pipeline through parallel computing (e.g., multiprocessing, GPU acceleration) to reduce runtime.
- ✓ **DEM Generation Module:** Integrate a complete DEM generation and visualization module using triangulation or dense stereo matching.
- ✓ **GUI Integration:** Develop a user-friendly interface to allow interactive image selection, parameter tuning, and visualization of results.

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